Yung Hsing (Winston) Huang

Bioinformatics/Cancer Research Expert

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SUMMARY/OBJECTIVE

- 7+ years of academic and industrial cancer research experience
- 2+ years of large-scale spatial transcriptomics experience
- Research expertise: Tumor microenvironment, spatial transcriptomics, single-cell mutliomics
- Solid background and experience in cancer biology and molecular biology techniques
- Lawful permanent resident in United States and a lawful citizen in Canada

WORK EXPERIENCE

Bioinformatician >> Senior Bioinformatician

Houston Methodist Research Institute | Houston, TX, USA | 04/2026 – Present

Spatial Omics Core, Houston Methodist Research Institute

Data analyses for Spatial Omics Core of Houston Methodist Research Institute including bulk-RNA, single-cell RNA (10X Chromium), spatial transcriptomics (10X Xenium) and spatial proteomics (Akoya Phenocycler) data, some key collaborations include:

- Integrating **Xenium** spatial transcriptomics and single-cell **T-cell repertoire sequencing** datasets from patients in our clinical collaborator's clinical trial (phase II, combined immune checkpoint inhibitor with radiation) to investigate the radiation vaccination effect on T cell clonality and immune-active niches in the tumor microenvironment of muscle-invasive bladder cancer patients (in collaboration with Dr. Raj Satkunasivam)
- Leveraging **Xenium** spatial transcriptomics to characterize the niche composition and essential ligand-receptor interactions of therapy-induced senescent tumor cells that are enriched in minimal residual cancers (Collaboration with Dr. Yulin Li).
- Employing in-house **Visium HD** datasets to assess cancer cell death mechanisms by comprehensively comparing the niches where tumor cells that are highly pyroptotic, senescent and/or necrotic by measures of three independent gene signatures derived by us and our collaborators (collaboration with Dr. Yulin Li and Dr. Yi-nan Gong)
- Applying **Xenium** spatial transcriptomics to characterize neighborhood corresponding to NK cell immaturity in triple-negative breast cancer (collaboration with Dr. Chaka
- Analysis of in-house bulk RNAseq dataset from patients with kidney clear cell carcinoma to study the role of thrombosis and cancer metastasis (collaboration with Dr. Dharam Kaushik)

Keith Syson Chan Lab, Department of Urology

I participate in the following projects as an in-lab bioinformatician by providing manpower in data mining, hypothesis validation, statistical analyses and exploratory data analysis:

- Harnessing in-house **single-cell RNAseq** and **Xenium 5k** spatial transcriptomics analysis to identify heterogeneity of cancer-associated fibroblasts (CAF) in non-muscle-invasive and muscle-invasive bladder cancer patients from multiple public datasets (https://doi.org/10.1126/sciadv.adt8697 and a manuscript in preparation)
- Development of Caspase-1 driven pyroptosis gene signature using Cox proportional hazard (CPH) model and utilizing in-house **Xenium** spatial transcriptomics to visualize the intratumoral pyroptotic niche that results in local increase of lymphocyte-to-neutrophil ratio (Wong et al., *Nature Communications*, *In Press*)
- Implementing gene signature scoring on public bulk and single-cell RNAseq datasets to validate the effect of extracellular collagen and tumor cell-expressing DDR1 on matrix-induced stress response and translation machinery (Alonzo et al. *Nature Communications, Under Revision*)
- Utilizing in-house **single-cell RNAseq** dataset prepared from patient-derived lymph node metastases to unpuzzle the fibroblast and endothelial subtypes that contribute to response of immunotherapy in bladder cancer (Results published at 2026 Keystone Symposium, Stromal Immunology by Dr. Grace L. Wong)
- Leveraging in-house **mouse single-cell RNAseq** dataset to study the effect of T cell-mediated anti-cancer immunity upon high fat diet-induced obesity in bladder cancer
- Applying in-house **mouse single-cell RNAseq** from intravenously inoculated bladder-to-lung metastasis tissue to study the relationship between B cells and NK cell activity in metastasis microenvironment.
- Employing in-house **Xenium 5k** spatial transcriptomics of non-muscle invasive bladder cancer to study the roles of tertiary lymphoid structure on disease progression.

Bioengineer

3D Systems | Houston, TX, USA | 06/2022 – 11/2023

I worked for a subsidiary company of 3D Systems, Systemic Bio, which focused on development of organ/tumor-on-chip development for the purposes of preclinical drug efficacy/toxicity assessment. My main job functions included:

- Assay development and optimization for samples collected from 3D organ/tumor-on-chip systems including viability/cytotoxicity assays (LDH, Cell-Titer Glo, MTS, CCK8), histology assays (cryosessioning of 3D-printed hydrogels, IHC/H&E staining), analyte quantification assays (ELISA of soluble proteins, bile acids, enzyme activity colorimetric assays) and nucleic acid assays (qPCR, PicoGreen).
  Assessment of biocompatibility of organ/tumor-on-chip systems using primary human hepatocytes and immortalized human cell lines using various biomaterials, microfludic systems and hydrogel print patterns
- Establishment of pipelines for biological/biochemical assay data collection, analyses, visualization and interpretation
  Maintenance of core experimental equipment including Leica Cryostat, MiniAmp™ Plus Thermal Cycler, Nanodrop, QuanStudio 6, MACSQuant 10 Analyzer Flow Cytometer and Promega Glomax Plate Reader
- Participation of partnership million-scale funding projects using organ/tumor-on-chip for preclinical drug screening with two major pharmaceutical companies.

Postdoctoral Fellow

Houston Methodist Research Institute | Houston, TX, USA | 04/2019 – 05/2022

- Identified an innovative therapeutic target NHE6 for improved chemotherapy and antibody-based immunotherapy efficacy in multiple myeloma patients using Oncomine and GEO data
- Genetic manipulation of NHE6 expression with siRNA, shRNA and CRISPR-Cas9 technology
- Data mining of potential target genes/signaling pathways that contribute to obesity-mediated myeloma tumorigenesis using in-house RNAseq data (GSE132604)
- Discovered the impact of retinoic acid receptor (RARϒ) and interferon β signaling pathway as a mechanism of carfilzomib resistance in multiple myeloma using in-house RNAseq data (GSE178340)
- Data mining of potential target genes/gene signaling pathways that are correlated with high-risk multiple myeloma using publicly available gene datasets on GEO.
- Patient survival analyses in myeloma patients segregated by gene expression level using publicly available GEO and Oncomine data
- Writing and preliminary data generation for a successfully funded CPRIT grant titled "Targeting NHE6 to improve clinical efficacy of daratumumab in myeloma" (Grant ID: RP220639)
- One manuscript titled "Inhibition of the NHE6-mediated endosome pathway improves therapeutic response in multiple myeloma" under revision (Nature Communication).

EDUCATION

Master of Science (M.Sc.) in Biomedical Informatics

University of Texas Health Science Center at Houston | Houston, TX, USA | Class of 2024

- Training on programming tools for biomedical informatics including MySQL, Python (pandas, numpy, matplotlib, seaborn, jupyter notebook etc), R (BiocManager and tidyverse), Google Colab, Tableau and Linux
- Practicum project (BMI 6000, Supervisor: Dr. Pora Kim):
  - Predict possible the breakpoints of KMT2A-AFF1 fusion proteins using the deep learning model FusionAI and FusionGDB2.0
  - Predict 3D protein structures of a fusion protein KMT2A-AFF1 using Alphafold2
  - Validate predicted structure using SAVES v6.0 and AlphaPickle
  - Prepare and identify potential active sites within a predicted protein structure for in silico drug binding screening using Schrodinger Maestro (Protein Preparation, SideMap, Grid and Glide)
  - Extract KMT2A-AFF1 breakpoint information from CCLE database using Linux
  - Perform RNAseq data alignment using STAR and Arriba to identify the presence of KMT2A-AFF1 fusion proteins in established AML/ALL cell lines
- Course project (BMI 5333 Systems Medicine: Principles and Practice): investigation on the genetic difference in black Americans compared to other ethnicities that could lead to worse survival and therapy response in myeloma patients using multiple myeloma research foundation (MMRF) database
- Course project (BMI 5330 Introduction to Bioinformatics): investigation on how obesity-related genomic defects could contribute to tumorigenesis using multiple bioinformatic databases including UCSC genome browser, dbSNP, ClinVAR, GEO, GWAS, 1000 Genome Project, TCGA, DAVID, BioGrid, OMIM etc

Doctor of Philosophy (Ph.D.) in Lab Medicine and Pathology

University of Alberta | Edmonton, AB, Canada | Class of 2019

- Development of a 3D cell culture model for therapy assessment and molecular analyses of primary myeloma cells and myeloma cell lines

- Development and preclinical assessment of anti-CD38-conjugated nanoparticles encapsulating STAT3 inhibitor as an innovative therapeutic agent for multiple myeloma

Bachelor of Science, Honours Degree (BSc.(Hon.)) in Biochemistry
University of British Columbia | Vancouver, BC, Canada | Class of 2013

PATENT

Pyroptosis-Associated Gene Signature (CASCADE) for Predicting Treatment Response in Bladder Cancer – Filed internally at Houston Methodist Research Institute # DISC2025014

RELEVANT CORE SKILLS

- Bioinformatics data processing and cleaning
- NGS analysis (bulk, scRNA, spatial omics)
- Bioinformatics databases (TCGA, GSE, etc)
- Version control and script management (GitHub)
- Pathway enrichment analyses (GO and KEGG, etc)
- Immune profile deconvolution analyses
- In silico simulation for drug discovery
- Data visualization and presentation
- Gene signature development
- Single-cell multiomics analyses
- Programming languages (Python and R)
- Linux for large-scale data manipulation

RESEARCH CONTRIBUTIONS

Peer-reviewed publications

1. Wong, S. Q., Hayashi, K., Subel, E., Gao, H., **Huang, Y. H.**, Nikolo, F., Karabicici, M., Hoi, X. P., Alonzo, M., Kasabyan, A., Tsouko, E., Bouchier-Hayes, L., Shin, C., Porter, S., Garcia, H., Yin, Z., Wong, S. T., El-Zaatari, Z., Satkunasivam, R., Chen, C., Brooks, M., Zhou, L., Korentzelos, D., Gong, Y., Li, Y., Titus, R. & Chan, KS. (2026). Chemotherapy-induced caspase-1 dependent pyroptosis in cancer cells remotely skews myelopoiesis to drive pro-tumorigenic systemic neutrophil-dominant inflammation. *Nat Commun (2026).* https://doi.org/10.1038/s41467-026-71471-3
2. Gao, H., Wong, S., Subel, E., **Huang, Y. H.**, Lee Y. C., Hayashi, K., Alonzo, M. E., Karabicici, M., Hoi, X. P., Kasabyan, A., Mo, Q., Zaatari, Z., Shen, S., Satkunasivam, R., Nikolos, F. & Chan, K. S. (2025). Caspase-1 dependent pyroptosis converts αSMA+ CAFs into collagen-IIIhigh iCAFs to fuel chemoresistant cancer stem cells. *Sci. Adv.11,eadt8697(2025).* https://doi.org/10.1126/sciadv.adt8697
3. **Huang, Y. H.**, Vakili, M. R., Molavi, O., Morrissey, Y., Wu, C., Paiva, I., Soleimani, A. H., Sanaee, F., Lavasanifar, A., & Lai, R. (2019). Decoration of anti-CD38 on nanoparticles carrying a STAT3 inhibitor can improve the therapeutic efficacy against myeloma. *Cancers*, *11*(2). https://doi.org/10.3390/cancers11020248
4. Alshareef, A., Zhang, H. F., **Huang, Y. H.**, Wu, C., Zhang, J. D., Wang, P., El-Sehemy, A., Fares, M., & Lai, R. (2016). The use of cellular thermal shift assay (CETSA) to study Crizotinib resistance in ALK-expressing human cancers. *Scientific Reports*, *6*(1), 1–12. https://doi.org/10.1038/srep33710
5. Haque, M., Li, J., **Huang, Y. H.**, Almowaled, M., Barger, C. J., Karpf, A. R., Wang, P., Chen, W., Turner, S. D., & Lai, R. (2019). Npm-alk is a key regulator of the oncoprotein foxm1 in alk-positive anaplastic large cell lymphoma. *Cancers*, *11*(8). https://doi.org/10.3390/cancers11081119
6. Liu, H., He, J., Bagheri-Yarmand, R., Li, Z., Liu, R., Wang, Z., Bach, D.-H., **Huang, Y. H.**, Lin, P., Guise, T. A., Gagel, R. F., & Yang, J. (2022). Osteocyte CIITA aggravates osteolytic bone lesions in myeloma. *Nature Communications*, *13*(1), 3684. https://doi.org/10.1038/s41467-022-31356-7
7. Saqr, A., Vakili, M. R., **Huang, Y. H.**, Lai, R., & Lavasanifar, A. (2019). Development of Traceable Rituximab-Modified PEO-Polyester Micelles by Postinsertion of PEG-phospholipids for Targeting of B-cell Lymphoma. *ACS Omega*, *4*(20), 18867–18879. https://doi.org/10.1021/acsomega.9b02910
8. Soleimani, A. H., Garg, S. M., Paiva, I. M., Vakili, M. R., Alshareef, A., **Huang, Y. H.**, Molavi, O., Lai, R., & Lavasanifar, A. (2017). Micellar nano-carriers for the delivery of STAT3 dimerization inhibitors to melanoma. *Drug Delivery and Translational Research*, *7*(4), 571–581. https://doi.org/10.1007/s13346-017-0369-4
9. Wang, P., Haque, M., Li, J., **Huang, Y.-H.**, Almowaled, M., Bargar, C., Karpf, A., Chen, W., Turner, S., & Lai, R. (2018). FOXM1 and the NPM-ALK/STAT3 Axis Form a Novel Positive Feedback Loop in Promoting the Oncogenesis of ALK-Positive Anaplastic Large Cell Lymphoma. *Blood*, *132*(Supplement 1), 3921–3921. https://doi.org/10.1182/blood-2018-99-115743
10. Wang, Q., Lin, Z., Wang, Z., Ye, L., Xian, M., Xiao, L., Su, P., Bi, E., **Huang, Y.-H.**, Qian, J., Liu, L., Ma, X., Yang, M., Xiong, W., Zu, Y., Pingali, S. R., Xu, B., & Yi, Q. (2021). RARγ activation sensitizes human myeloma cells to carfilzomib treatment through OAS-RNase L innate immune pathway. *Blood.* https://doi.org/10.1182/blood.2020009856
11. Wang, Z., He, J., Bach, Dh., **Huang, Y.-H.** et al. Induction of m6A methylation in adipocyte exosomal LncRNAs mediates myeloma drug resistance. J Exp Clin Cancer Res 41, 4 (2022). https://doi.org/10.1186/s13046-021-02209-w
12. Wu, C., Gupta, N., **Huang, Y. H.**, Zhang, H. F., Alshareef, A., Chow, A., & Lai, R. (2018). Oxidative stress

enhances tumorigenicity and stem-like features via the activation of the Wnt/β-catenin/MYC/Sox2 axis in ALK-positive anaplastic large-cell lymphoma. *BMC Cancer*, *18*(1), 1–11. https://doi.org/10.1186/s12885-018-4300-2
13. Wu, C., Zhang, H. F., Gupta, N., Alshareef, A., Wang, Q., **Huang, Y. H.**, Lewis, J. T., Douglas, D. N., Kneteman, N. M., & Lai, R. (2016). A positive feedback loop involving the Wnt/β-catenin/MYC/Sox2 axis defines a highly tumorigenic cell subpopulation in ALK-positive anaplastic large cell lymphoma. *Journal of Hematology and Oncology*, *9*(1), 1–14. https://doi.org/10.1186/s13045-016-0349-z

Theses

1. Doctoral Thesis - **Huang, Y. H.** (2019). Unmasking the in vivo importance of STAT3 in multiple myeloma using three-dimensional culture and nanoparticle drug delivery system.
2. Undergraduate Honours Thesis - **Huang, Y. H.** (2013). Intracellular Distribution of Acetyl-CoA Carboxylase in HepG2 Cells and Effects of Phosphorylation.

Conferences/Presentations:

1. **Huang Y. H.** Xenium Spatial Analysis Identifies Outcome-linked Fibroblast Subpopulations in NMIBC. Oral Presentation at Houston Methodist Research Institute seminar, Houston, TX, USA. Apr 2025.
2. **Huang Y. H.**, Molavi O, Alshareef A, Haque M, Wang Q, Chu MP, Venner CP, Peters AC, Lavasanifar A and Lai R. Constitutive activation of STAT3 in myeloma cells cultured in a three-dimensional, reconstructed bone marrow model. Poster presented at the meeting of 2017 Canadian Cancer Research Conference (CCRC), Vancouver, Canada. Event website: https://www.ccra-acrc.ca. Nov 2017.
3. **Huang, Y. H.**, Haque, M., Lavasanifar, A., and Lai, R. Myeloma cells acquire STAT3 activity and dependence in established three-dimensional reconstructed bone matrix (3D/rBM) model. Poster presented at the Discovery, Research, Innovative and Education (DRIvE) Day, Unversity of Alberta, Edmonton, Canada. Apr 2017.
4. **Huang, Y. H.**, Lavasanifar, A., and Lai, R. STAT3 Inhibitor Packaged in Nanoparticles as a Strategy for Its Better Delivery in STAT3-Targeting Cancer Therapy. Oral presentation at Discovery, Research, Innovative and Education (DRIvE) Day, Unversity of Alberta, Edmonton, Canada. Apr 2016.
5. **Huang, Y. H.**, Soleimani, A., Wu, CS., Morrissey, Y., Lavasanifar, A., and Lai, R. STAT3 inhibitor packaged in nanoparticle as a strategy for its better delivery in STAT3-targeting cancer therapy. Poster presented at the meeting of 2015 Canadian Cancer Research Conference (CCRC), Montreal, Canada. Event website: https://www.ccra-acrc.ca. Nov 2015.
6. **Huang, Y. H.**, Soleimani, A., Wu, CS., Morrissey, Y., Lavasanifar, A., and Lai, R. Encapsulation of an anti-STAT3 agent for enhanced delivery and improved safety. Poster presented at the annual meeting of Cancer Research in North Alberta Conference (CRINA), Edmonton, Canada. Abstract listed at: https://uofa.ualberta.ca/cancer-institute/-/media/crina/events/research-day-2015-abstract-booklet.pdf (poster# 51). Nov 2015.
7. Paiva, I., Soudy, R., Soleimani, A., **Huang, Y. H.**, Kamaljit, K., & Lavasanifar, A. (2017). Near-infrared optical imaging of c18-DK decapeptide in an orthotopic breast cancer mouse model. *2017 Controlled Release Society Annual Meeting*.